

# FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

## FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 1.2 INDICES: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Why do scientists, economists and farmers use small superscript numbers? Look back at the two cards from Lesson 1: Card A shows Kenya's 2019 Census figure 47,564,296 written in full, and Card B shows  $4.8 \times 10^7$ . Answer every section in the space under the prompt. Write in full sentences, show your working line by line, and name the law of indices you use at each step. Use the ideas from all eight lessons: repeated multiplication, the product law, the quotient law, the zero-index law and the power-of-a-power law. Where a question asks you to explain, give a reason, not just an answer.

### Part 1: Reading the small superscript number

**Look again at Card B:  $4.8 \times 10^7$ . (a) Name the base and the index (power) in  $10^7$ , and say what each one tells you. (b) Write  $10^7$  as repeated multiplication and give its value in ordinary form. (c) Explain why Card B uses a superscript 7 when Card A is the Census figure 47,564,296. Why is Card B shorter, easier to compare and less likely to be misread? (d) Write  $2^5$  as repeated multiplication and evaluate it.**

(a) In  $10^7$  the base is 10, the number being multiplied by itself. The index (power or exponent) is 7, which tells us how many times the base is multiplied by itself. (b)  $10^7 = 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 10,000,000$  (ten million). (c) The Census figure is about 4.8 ten-millions, so we write  $4.8 \times 10^7$ . The superscript 7 replaces seven separate multiplication signs (or the long place-value string of ten million) with one small symbol. Card A has eight digits and two commas, while Card B has a short number and one small superscript. Card B is easier to compare with other numbers, easier to compute with and less likely to be misread, because the superscript tells you the size directly and you do not have to count digits. Card B is an approximation: 47,564,296 is about  $4.76 \times 10^7$ , and  $4.8 \times 10^7$  is that value rounded to 2 significant figures, so it keeps almost all of the information. (d)  $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$ .

### Part 2: The product law and projecting the population

**(a) Expand  $10^2 \times 10^3$  fully as repeated multiplication, count all the factors, and write the answer as a single power of 10. (b) Use your working to explain in words why multiplying powers with the same base means adding the indices. Write**

(a)  $10^2 \times 10^3 = (10 \times 10) \times (10 \times 10 \times 10) = 10 \times 10 \times 10 \times 10 \times 10$ , which is 5 factors of 10, so the answer is  $10^5$ . (b) The first power puts 2 copies of the base into the product and the second puts 3 copies. Together there are  $2 + 3 = 5$  identical factors, so the indices are added. In general  $a^m \times a^n = a^{m+n}$ , because  $m$  copies and  $n$  copies of  $a$  make  $m + n$  copies. (c)  $4.8 \times 10^7 \times 10^2 = 4.8 \times 10^{7+2} = 4.8 \times 10^9$ . This is the projection from the rounded figure  $4.8 \times 10^7$ ; the exact Census figure 47,564,296 would give 4,756,429,600, which is about  $4.76 \times 10^9$  and rounds to  $4.8 \times 10^9$  to 2 significant figures. We only added the small indices 7 and 2 instead of multiplying out a long number and writing a ten-digit answer. This is quicker, and it is less likely to produce a mistake with the zeros.

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| <p>the general law using <math>a</math>, <math>m</math> and <math>n</math>. (c) Suppose Kenya's population of about <math>4.8 \times 10^7</math> were projected to grow by a factor of <math>10^2</math>. Use the product law to write the projected population in the form <math>a \times 10^n</math>. Explain why this is quicker than multiplying out the full numbers.</p> |  |
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| Part 3: Division and the surprising power zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>(a) Expand <math>2^5 \div 2^2</math> as a fraction, cancel the common factors, and write the answer as a single power. State the quotient law for <math>a^m \div a^n</math>. (b) Now work out <math>2^3 \div 2^3</math> in two ways: first by the quotient law, then by ordinary arithmetic. What do the two results tell you about <math>2^0</math>? (c) A classmate says: 'Zero copies of a number must be zero, so <math>a^0 = 0</math>.' Use your results from (b) to explain whether your classmate is right.</p> | <p>(a) <math>2^5 \div 2^2 = (2 \times 2 \times 2 \times 2 \times 2) / (2 \times 2)</math>. Cancelling two 2s from the top and bottom leaves <math>2 \times 2 \times 2 = 2^3</math>. The quotient law is <math>a^m \div a^n = a^{m-n}</math>, because <math>n</math> of the <math>m</math> factors cancel and <math>m - n</math> remain. (b) By the quotient law, <math>2^3 \div 2^3 = 2^{3-3} = 2^0</math>. By arithmetic, <math>8 \div 8 = 1</math>. Both are correct, so <math>2^0 = 1</math>. (c) Your classmate is not right. Our results show that <math>2^0 = 1</math>, not 0, because <math>2^3 \div 2^3</math> equals 1 by arithmetic and equals <math>2^0</math> by the quotient law. The same works for any non-zero number: dividing it by itself gives 1, and the quotient law says the same division gives <math>a^0</math>, so <math>a^0 = 1</math> for any <math>a \neq 0</math>. When all the factors cancel, 1 is what remains, not 0.</p> |

| Part 4: Power of a power and multi-step problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <p>(a) Expand <math>(2^3)^2</math> fully, count the factors of 2, and write the result as a single power. State the law for <math>(a^m)^n</math>. (b) In computer memory, 1 GB = <math>(2^{10})^3</math> bytes and 1 TB = <math>(2^{10})^4</math> bytes. Use your law to write 1 TB as a single power of 2. (c) Simplify <math>(3^2 \times 3^5) \div 3^4</math> and evaluate your answer. List the law you use at each step. (d) Simplify <math>((2^3)^2 \times 2^4) \div 2^6</math> and evaluate your answer. Explain why you must deal with the bracket first.</p> | <p>(a) <math>(2^3)^2 = 2^3 \times 2^3 = (2 \times 2 \times 2) \times (2 \times 2 \times 2)</math>, which is 6 factors of 2, so <math>(2^3)^2 = 2^6</math>. The law is <math>(a^m)^n = a^{mn}</math>, because <math>n</math> copies of <math>a^m</math> give <math>m \times n</math> copies of <math>a</math>. (b) 1 TB = <math>(2^{10})^4 = 2^{10 \times 4} = 2^{40}</math> by the power-of-a-power law (multiply the indices). (c) Step 1, product law: <math>3^2 \times 3^5 = 3^7</math>. Step 2, quotient law: <math>3^7 \div 3^4 = 3^3</math>. <math>3^3 = 27</math>. (d) Step 1, power of a power: <math>(2^3)^2 = 2^6</math>. Step 2, product law: <math>2^6 \times 2^4 = 2^{10}</math>. Step 3, quotient law: <math>2^{10} \div 2^6 = 2^4 = 16</math>. The bracket comes first because the power-of-a-power law must be resolved before the product or quotient law can combine the powers into a single base and index.</p> |

## Part 5: Answering the Driving Question

**Driving Question: Why do scientists, economists and farmers use small superscript numbers to write huge or tiny quantities, and what rules make those compact symbols so powerful for computation? Write a paragraph of 6 to 9 sentences to explain Card B ( $4.8 \times 10^7$ ) to a younger learner who has never seen index notation. Your paragraph must (i) explain what the 7 means, (ii) name all four laws of indices and say what each does to the indices, (iii) explain the single big idea that all four laws come from, and (iv) give one real-life Kenyan application of index laws, for example from census data, the national budget, mobile data storage or M-Pesa.**

The small 7 in  $4.8 \times 10^7$  tells us that 10 is multiplied by itself 7 times, giving ten million, and the Census figure 47,564,296 is about 4.8 of those ten-millions, rounded to 2 significant figures. The index follows four neat rules: the product law adds indices ( $a^m \times a^n = a^{m+n}$ ), the quotient law subtracts them ( $a^m \div a^n = a^{m-n}$ ), the power-of-a-power law multiplies them ( $(a^m)^n = a^{mn}$ ), and the zero-index law says  $a^0 = 1$  for any  $a \neq 0$ . All four laws come from one big idea, repeated multiplication: when we count or cancel the factors we get the rules, so they are not arbitrary tricks. For example, if Kenya's population of  $4.8 \times 10^7$  grew by a factor of  $10^2$ , we only add  $7 + 2$  to get about  $4.8 \times 10^9$ . A government planner or ICT technician can compute with huge numbers, such as the national budget of about  $3.7 \times 10^{12}$  shillings or the storage of 1 TB =  $2^{40}$  bytes, without writing out every digit. Indices matter because they make working with very large and very small quantities compact, exact and efficient.

## RUBRIC

| Criterion                              | Excellent (4)                                                                                                                                                                           | Proficient (3)                                                                                                     | Developing (1–2)                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Meaning of index notation              | Correctly names base and index, expands powers as repeated multiplication with correct values, and clearly explains why Card B is more compact and less error-prone than Card A.        | Names base and index and expands correctly, with only a brief or partial explanation of why index form is compact. | Confuses base and index, or expands incorrectly, or gives no explanation of the compact form.                       |
| Applying the four laws accurately      | Applies the product, quotient, zero-index and power-of-a-power laws correctly in every part, including the multi-step problems, with the right order of steps and correct final values. | Applies most laws correctly with minor slips in the multi-step problems or in evaluating.                          | Applies laws incorrectly, for example multiplying indices in the product law, or cannot combine laws in a sequence. |
| Reasoning from repeated multiplication | Justifies each law by expanding and counting or cancelling factors, and explains why $a^0 = 1$ using both the                                                                           | Justifies some laws by expansion but states others without a reason.                                               | States the laws as rules to memorise with no justification, or believes $a^0 = 0$ .                                 |

|                                  |                                                                                                                                                                             |                                                                                                 |                                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                                  | quotient law and arithmetic.                                                                                                                                                |                                                                                                 |                                                                                                 |
| Communication and Kenyan context | Writes clearly, shows each line of working, names the law used at each step, and links the ideas accurately to the Census figure and at least one other Kenyan application. | Working is mostly clear and the laws are named, with a general or partly developed Kenyan link. | Working is missing or hard to follow, laws are not named, and there is no relevant Kenyan link. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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